

SOP 201: MRI Safety Screen

1 Summary

All individuals must be screened before they enter the Magnet Room. This document establishes MRI Safety screening procedures at the Notre Dame Human Neuroimaging Center (ND-HNC).

2 Scope

These policies apply to all individuals who enter Zones 3 & 4 (B15, B17, B17a; Fig. 1).

3 Definitions

- **Magnet Room:** B17A; The room within the MRI suite that contains the MRI scanner. It is accessed from within the MRI Control room (B17) through a doorway that is controlled by an electronic lock.
- **ND-HNC:** Notre Dame Human Neuroimaging Center, Lower Level of Veldman Family Psychology Clinic.
- **Operator:** MRI Technologist or Level 4 researcher, responsible for conducting research operations and maintaining equipment. Typically the Lead during emergency responses.
- **Visitor:** An individual who has not taken the ND-HNC MRI Safety Course but has a legitimate reason for entering the MRI suites. Research participants and their legal guardians are Visitors. Other Visitors may include: individuals providing/receiving a tour of the VFPC and ND-HNC, faculty and staff who have not received MRI Safety Course, and research assistants.

4 Notes

- Emergency response and maintenance personnel **CANNOT** enter the Magnet Room unless they have been screened and all metal objects have been removed.
- Emergency response personnel can **ONLY** enter Zone 3 without completing an MRI Safety screener if the door to the Magnet Room has been locked by key.
- ND-HNC only stores MRI Safety forms for staff and researchers. Visitor forms are maintained by their respective research teams.

5 Policies & Procedures

The purpose of MRI Safety screening is to determine whether entry into the Magnet Room poses a health or safety risk. Accordingly, all individuals entering Zones 3 & 4 must be screened according to the procedures below.

5.1 MRI Prescreen

Prescreening is conducted for Visitor's intending to enter the Magnet Room (e.g. a research participant). Before arriving at the ND-HNC, research personnel check for potential MRI contraindications with the prospective Visitor:

- Non-MRI compatible medical implants.
 - **Note:** Visitors with an implant must present documentation from their surgeon's office that the device is MRI compatible.
 - **IUDs:** North American non-metallic intrauterine devices (e.g. Mirena) are approved. Metallic (e.g. Cu) and IUDs from other regions must be approved by the MRI Technologist.
- Other metal embedded in the body (e.g. shrapnel, unremovable piercings).
- Pregnancy.
- Claustrophobia (the Mock can be used to assess comfort levels).
- Tattoo on face or neck, including permanent makeup.

5.2 MRI Preparation

Visitors intending to enter the Magnet Room (e.g. a research participant) should be informed of the following restrictions, before arriving at the ND-HNC, to guide their preparations:

- Clothing should be free of metallic objects and clasps; bras should not have an underwire. Anti-odorant materials (e.g. Lululemon's Silverescent) are not MRI compatible.
- Jewelry will need to be removed, including piercings if they are metallic.
- Cosmetics such as colored contacts, magnetic lashes, and makeup or nail polish with glitter will have to be removed. Tanning lotions are not MRI compatible.
- Hair must be free of all accessories (metallic clips, barrettes, scrunchies, etc.).
- If Visitors have corrected vision, they should wear contacts or know their prescription.

5.3 MRI Safety Screen

All who enter Zone 3, regardless of whether they will enter Zone 4 (Magnet Room) fill out Form 001: MRI Safety. ND-HNC staff and research personnel submit a form (a) annually, and (b) whenever a change occurs in their MRI compatibility. Visitors fill out the form in Zone 1, and researchers with at least Level 2 access should be available to answer any questions.

- For Visitors, Form 001 must be filled out on the same day as entering Zone 3.
- Each question/prompt must be answered individually.
- In the case of minors, their legal guardian must fill out the form.
 - Note: verify guardianship, which may differ from the individual who brought the minor to the center.
- The Visitor signs and dates the form in the designated location.

5.4 MRI Safety Screen, Review 1

A researcher with at least Level 2 access will review the Visitor's responses on Form 001.

- Discuss any potential contraindications, points of confusion, or concerns. Inform the Operator about any items discussed or addressed.
- After reviewing Form 001, the researcher prints and signs their name in the designated location.
- If the researcher determines the Visitor is MRI compatible, they may escort the Visitor into Zones 2 & 3.

5.5 MRI Safety Screen, Review 2

A second review is required for Visitors entering Zone 4 (Magnet Room).

- The Operator conducts the second review, who must not be the same person that conducted Review 1.
- Any potential contraindications, points of confusion, or concerns are addressed, including those raised by the first reviewer
- The Operator signs and dates Form 001.
- The Visitor may enter Zone 4 only after the Operator has determined that it is safe to do so.

